Practical no.06 (OS)

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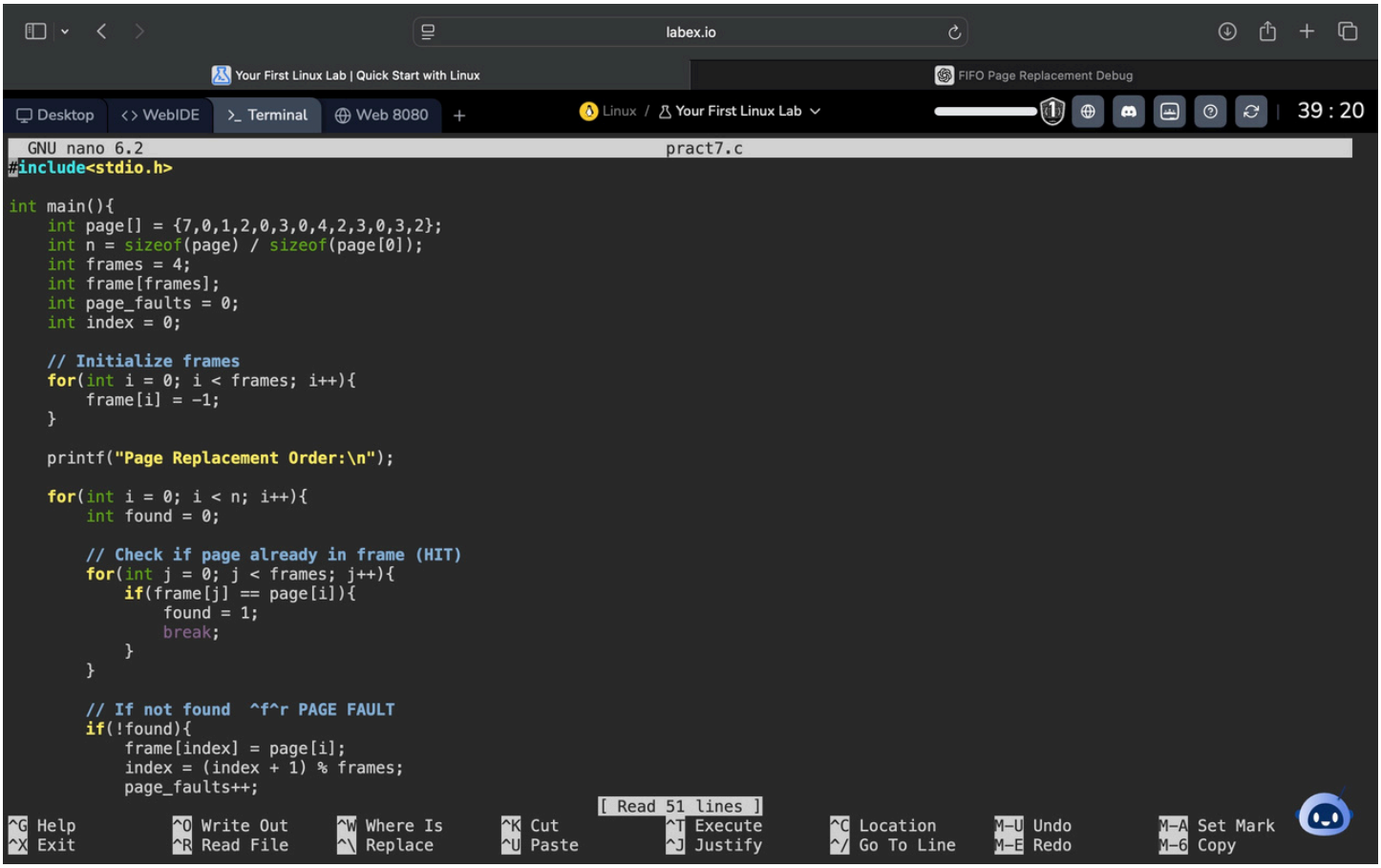
SECTION:A7_B3_61.

1)FIFO




```c
#include<stdio.h>

int main(){
    int page[] = {7,0,1,2,0,3,0,4,2,3,0,3,2};
    int n = sizeof(page) / sizeof(page[0]);
    int frames = 4;
    int frame[frames];
    int page_faults = 0;
    int index = 0;

    // Initialize frames
    for(int i = 0; i < frames; i++){
        frame[i] = -1;
    }

    printf("Page Replacement Order:\n");

    for(int i = 0; i < n; i++){
        int found = 0;

        // Check if page already in frame (HIT)
        for(int j = 0; j < frames; j++){
            if(frame[j] == page[i]){
                found = 1;
                break;
            }
        }

        // If not found  ^f^r PAGE FAULT
        if(!found){
            frame[index] = page[i];
            index = (index + 1) % frames;
            page_faults++;
```

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FIFO Page Replacement Debug

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GNU nano 6.2 pract7.c

```
        // Check if page already in frame (HIT)
        for(int j = 0; j < frames; j++){
            if(frame[j] == page[i]){
                found = 1;
                break;
            }
        }

        // If not found  ^f^r PAGE FAULT
        if(!found){
            frame[index] = page[i];
            index = (index + 1) % frames;
            page_faults++;
        }

        // Print frames
        printf("Page %d: ", page[i]);
        for(int j = 0; j < frames; j++){
            if(frame[j] != -1){
                printf("%d ", frame[j]);
            } else {
                printf("- ");
            }
        }
        printf("\n");
    }

    printf("Total Page Faults: %d\n", page_faults);

    return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

OUTPUT:

```
labex:A7_B3_61/ $ ./a.out
Page Replacement Order:
Page 7: 7 - - -
Page 0: 7 0 - -
Page 1: 7 0 1 -
Page 2: 7 0 1 2
Page 0: 7 0 1 2
Page 3: 3 0 1 2
Page 0: 3 0 1 2
Page 4: 3 4 1 2
Page 2: 3 4 1 2
Page 3: 3 4 1 2
Page 0: 3 4 0 2
Page 3: 3 4 0 2
Page 2: 3 4 0 2
Total Page Faults: 7
labex:A7_B3_61/ $
```

2)LRU

CODE:

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```
  GNU nano 6.2                                                    LRU.c
#include <stdio.h>

int main() {
    int pages[] = {1, 2, 1, 0, 3, 0, 4, 2, 4};
    int n = sizeof(pages) / sizeof(pages[0]);
    int frames = 3;

    int frame[3], time[3];
    int count = 0, faults = 0;

    // Initialize frames
    for(int i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    printf("Page\tFrames\t\tResult\n");

    for(int i = 0; i < n; i++) {
        int found = 0;

        // Check for HIT
        for(int j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                count++;
                time[j] = count;  // update recent use
                break;
            }
        }

        // If FAULT
        if(!found) {
                                    [ Read 68 lines ]
^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

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```
  GNU nano 6.2                                                    LRU.c
        // If FAULT
        if(!found) {
            int pos = 0;

            // Find least recently used
            for(int j = 1; j < frames; j++) {
                if(time[j] < time[pos]) {
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            count++;
            time[pos] = count;
            faults++;
        }

        // Print frames
        printf("%d\t", pages[i]);
        for(int j = 0; j < frames; j++) {
            if(frame[j] != -1)
                printf("%d ", frame[j]);
            else
                printf("- ");
        }

        if(found)
            printf("\tHit\n");
        else
            printf("\tFault\n");
    }

^G Help      ^O Write Out   ^W Where Is   ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace    ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

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GNU nano 6.2 LRU.c

```
            for(int j = 1; j < frames; j++) {
                if(time[j] < time[pos]) {
                    pos = j;
                }
            }

            frame[pos] = pages[i];
            count++;
            time[pos] = count;
            faults++;
        }

        // Print frames
        printf("%d\t", pages[i]);
        for(int j = 0; j < frames; j++) {
            if(frame[j] != -1)
                printf("%d ", frame[j]);
            else
                printf("- ");
        }

        if(found)
            printf("\tHit\n");
        else
            printf("\tFault\n");
    }

    printf("\nTotal Page Faults = %d\n", faults);
    printf("Total Page Hits = %d\n", n - faults);

    return 0;
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

OUTPUT:

```
labex:A7_B3_61/ $ nano LRU.c
labex:A7_B3_61/ $ gcc LRU.c
labex:A7_B3_61/ $ ./a.out
Page     Frames            Result
1        1 - -   Fault
2        1 2 -   Fault
1        1 2 -   Hit
0        1 2 0   Fault
3        1 3 0   Fault
0        1 3 0   Hit
4        4 3 0   Fault
2        4 2 0   Fault
4        4 2 0   Hit

Total Page Faults = 6
Total Page Hits = 3
labex:A7_B3_61/ $
```